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vboxuser@ubuntu:~/smodule/task3
~/smodule/task3

vboxuser@ubuntu:~/smodule/task3$ make
make -C /lib/modules/7.0.0-15-generic/build M=/home/vboxuser/smodule/task3 modules
make[1]: x86_64 katanora +/usr/src/linux-headers-7.0.0-15-generic
make[2]: x86_64 katanora +/home/vboxuser/smodule/task3
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc (Ubuntu 15.2.0-10ubuntu1) 15.2.0
You are using: gcc (Ubuntu 15.2.0-10ubuntu1) 15.2.0
warning: pahole version differs from the one used to build the kernel
The kernel was built with: 131
You are using: 0
make[2]: x86_64 katanora +/home/vboxuser/smodule/task3
make[1]: x86_64 katanora +/usr/src/linux-headers-7.0.0-15-generic
vboxuser@ubuntu:~/smodule/task3$ ls -l kbleds.ko
-rw-rw-r-- 1 vboxuser vboxuser 396080 Aug 25 17:17 kbleds.ko
vboxuser@ubuntu:~/smodule/task3$ sudo insmod ./kbleds.ko
[sudo: authenticate] Наполеон:
insmod: ERROR: could not insert module ./kbleds.ko: File exists
vboxuser@ubuntu:~/smodule/task3$ lsmod | grep kbleds
kbleds                12288  0
vboxuser@ubuntu:~/smodule/task3$ sudo dmesg | tail -n 20
[19372.008508] audit: type=1400 audit(1707647807.819:321): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="snap.firefox.hook.post-refresh" pid=10853 comm="apparmor_
parser"
[19372.003681] audit: type=1400 audit(1707647807.823:322): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="snap.update-ns.mesa-2404" pid=10854 comm="apparmor_parser"
[19372.007352] audit: type=1400 audit(1707647807.825:323): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="snap.mesa-2404.component-monitor" pid=10855 comm="apparmo
r_parser"
[19415.078177] kauditd_printk_skb: 4 callbacks suppressed
[19415.078183] audit: type=1326 audit(1707647134.392:328): auid=1000 uid=1000 gid=1000 ses=3 subj=snap.firefox.firefox pid=11245 comm=4267494F546872E6F6C28233131 exe="/snap/firefox/8763/usr/lib/firefox/firefox" sig=0 arch=C0000
0be syscall=251 event=1 ref=60a028d code=0x50000
[19507.452712] proc_module: loaded
[19507.452719] proc_module: /proc/proc created
[19536.799340] audit: type=1326 audit(1707647265.489:329): auid=1000 uid=1000 gid=1000 ses=3 subj=snap.firefox.firefox pid=11245 comm=4267494F546872E6F6C282337 exe="/snap/firefox/8763/usr/lib/firefox/firefox" sig=0 arch=C0000
0be syscall=251 event=1 ref=60a028d code=0x50000
[19551.439136] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 4 times, consider switching to WQ_UNBOUND
[19576.052218] workqueue: blk_mq_request_work hogged CPU for >10000us 5 times, consider switching to WQ_UNBOUND
[19610.311782] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 3 times, consider switching to WQ_UNBOUND
[19918.054265] proc_module: received from userspace: Hello from userspace
[21072.311911] proc_module: unloaded
[24119.907366] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 67 times, consider switching to WQ_UNBOUND
[25074.123278] kbleds: loading
[25074.123289] kbleds: loaded successfully
[25074.123390] kbleds: sysfs = /sys/kernel/kbleds/mask
[25423.109142] workqueue: blk_mq_request_work hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[27266.460806] audit: type=1400 audit(1707655639.544:330): apparmor="DENIED" operation="open" class="file" profile="snap.fireware-updater.fireware-notifier" name="/proc/sys/vm/max_map_count" pid=16652 comm="firmware-notifi" reques
ted_mask="r" denied_mask="r" fsuid=1000 ouid=0
[30778.726952] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 5 times, consider switching to WQ_UNBOUND
vboxuser@ubuntu:~/smodule/task3$ ls -l /sys/kernel/kbleds/
total 0
-rw-rw---- 1 root root 4096 Aug 25 19:17 mask
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[19610.311782] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 3 times, consider switching to WQ_UNBOUND
[19918.054265] proc_module: received from userspace: Hello from userspace
[21072.311911] proc_module: unloaded
[24119.907366] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 67 times, consider switching to WQ_UNBOUND
[25074.123278] kbleds: loading
[25074.123289] kbleds: loaded successfully
[25074.123390] kbleds: sysfs = /sys/kernel/kbleds/mask
[25423.109142] workqueue: blk_mq_request_work hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[27266.460806] audit: type=1400 audit(1707655639.544:330): apparmor="DENIED" operation="open" class="file" profile="snap.fireware-updater.fireware-notifier" name="/proc/sys/vm/max_map_count" pid=16652 comm="firmware-notifi" reques
ted_mask="r" denied_mask="r" fsuid=1000 ouid=0
[30778.726952] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 5 times, consider switching to WQ_UNBOUND
vboxuser@ubuntu:~/smodule/task3$ ls -l /sys/kernel/kbleds/
total 0
-rw-rw---- 1 root root 4096 Aug 25 19:17 mask
vboxuser@ubuntu:~/smodule/task3$ sudo cat /sys/kernel/kbleds/mask
0
vboxuser@ubuntu:~/smodule/task3$ echo 1 | sudo tee /sys/kernel/kbleds/mask
1
vboxuser@ubuntu:~/smodule/task3$ sudo cat /sys/kernel/kbleds/mask
1
vboxuser@ubuntu:~/smodule/task3$ echo 2 | sudo tee /sys/kernel/kbleds/mask
2
vboxuser@ubuntu:~/smodule/task3$ sudo cat /sys/kernel/kbleds/mask
2
vboxuser@ubuntu:~/smodule/task3$ echo 4 | sudo tee /sys/kernel/kbleds/mask
4
vboxuser@ubuntu:~/smodule/task3$ sudo cat /sys/kernel/kbleds/mask
4
vboxuser@ubuntu:~/smodule/task3$ echo 5 | sudo tee /sys/kernel/kbleds/mask
5
vboxuser@ubuntu:~/smodule/task3$ sudo cat /sys/kernel/kbleds/mask
5
vboxuser@ubuntu:~/smodule/task3$ echo 7 | sudo tee /sys/kernel/kbleds/mask
7
vboxuser@ubuntu:~/smodule/task3$ sudo cat /sys/kernel/kbleds/mask
7
vboxuser@ubuntu:~/smodule/task3$ echo 0 | sudo tee /sys/kernel/kbleds/mask
0
vboxuser@ubuntu:~/smodule/task3$ sudo cat /sys/kernel/kbleds/mask
0
vboxuser@ubuntu:~/smodule/task3$ sudo rmmod kbleds
vboxuser@ubuntu:~/smodule/task3$ lsmod | grep kbleds
vboxuser@ubuntu:~/smodule/task3$ sudo dmesg | tail -n 10
[21072.311911] proc_module: unloaded
[24119.907366] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 67 times, consider switching to WQ_UNBOUND
[25074.123278] kbleds: loading
[25074.123289] kbleds: loaded successfully
[25074.123390] kbleds: sysfs = /sys/kernel/kbleds/mask
[25423.109142] workqueue: blk_mq_request_work hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[27266.460806] audit: type=1400 audit(1707655639.544:330): apparmor="DENIED" operation="open" class="file" profile="snap.fireware-updater.fireware-notifier" name="/proc/sys/vm/max_map_count" pid=16652 comm="firmware-notifi" reques
ted_mask="r" denied_mask="r" fsuid=1000 ouid=0
[30778.726952] workqueue: e1000_watchdog [e1000] hogged CPU for >10000us 5 times, consider switching to WQ_UNBOUND
[32083.426060] kbleds: unloaded
[32863.426072] kbleds: unloaded
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